PROPLED: PROPERTY LEDGER

A PROJECT REPORT

Submitted by

NAMRATHA S R 201401032

VARUNESH R V 201401055

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RAJALAKSHMI ENGINEERING COLLEGE

ANNA UNIVERSITY: CHENNAI 600 025

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RAJALAKSHMI ENGINEERING COLLEGE

CHENNAI

BONAFIDE CERTIFICATE

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SIGNATURE

Dr. K. Devaki, M.E., Ph.D.,

PROFESSOR & HEAD OF THE DEPARTMENT

Department of Computer Science and Business Systems

Rajalakshmi Engineering College
Chennai – 602 105

SIGNATURE

Dr. P. Shanmugapriya, M.E, Ph.D.,

SUPERVISOR

PROFESSOR

Department of Computer Science and Business Systems

Rajalakshmi Engineering College
Chennai – 602 105

Submitted to Project Viva-Voce Examination held on _____

Internal Examiner **External Examiner**

ABSTRACT

The project aims to develop a user-friendly online platform that leverages database technology to efficiently display comprehensive property details to users. Administrators have the capability to oversee property listings, update tax information and historical owners' details of properties. Users are provided with a user-friendly interface to browse property listings, view tax dues, historical owners' data and initiate booking requests. By utilizing database management, the system will ensure scalability, data security, and real-time updates. The system's architecture incorporates backend services for data management and business logic, a frontend interface for user interaction, and a robust database system for data storage and retrieval. The integration of modern web technologies will facilitate easy navigation and searching, enabling users to retrieve accurate property information swiftly. Through this project, users will have a convenient and reliable tool to access property details, enhancing transparency and knowledge in the realm of real estate ownership.

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NAMRATHA S R

VARUNESH R V

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CHAPTER 1

INTRODUCTION

1.1 GENERAL

In the digital era, the real estate industry is witnessing a paradigm shift towards online platforms for property transactions and management. With the aim of revolutionizing this domain, the PROPLED (Property Ledger) project emerges as a comprehensive web application designed to streamline property transactions, enhance transparency, and empower users in their quest for buying, selling, and managing real estate assets

PROPLED serves as a centralized platform where users can seamlessly navigate through a myriad of features tailored to meet their diverse needs. Admins wield the power to authenticate users, ensuring a secure environment, while approved users gain access to a plethora of functionalities, including property listing, tax disclosure, legal dispute resolution, and transparent ownership history.

This project envisions a future where property transactions are simplified, transparent, and accessible to all stakeholders. By leveraging cutting-edge technologies and adhering to industry best practices, PROPLED aims to redefine the landscape of real estate management, fostering trust, efficiency, and innovation in every interaction.

The propose property booking system aims to simplify and streamline the process while enhancing transparency and security. It includes feature to verify tax and old owners details, allow multiple partners to be associated with one property, offer a user-friendly interface, and efficient data management tools. The system will facilitate communication between users and the admin. The goal is to create a fair and equitable allocation process that enhances the user experience. The proposed property booking system aims to simplify and streamline the process while enhancing transparency and security. It includes features to verify tax and old owners details, allow multiple partners to be associated with one property, offer a user-friendly interface, and efficient data management tools. The system will facilitate communication between users and the admin. The goal is to create a fair and equitable allocation process that enhancers the user experience.

In this report, we delve into the intricate details of the PROPLED project, exploring its key features, technological underpinnings, user experience considerations, and the broader implications for the real estate industry. Through meticulous analysis and comprehensive

evaluation, we aim to shed light on the transformative potential of PROPLED and its role in shaping the future of property management and transactions.

1.2 BACKGROUND

The real estate industry has traditionally been characterized by complex, time-consuming, and often opaque processes for property transactions and management. From cumbersome paperwork to lack of transparency in property listings, buyers and sellers have long grappled with inefficiencies and uncertainties in navigating the real estate market. Moreover, the prevalence of tax disputes, legal ambiguities, and incomplete ownership histories further exacerbates the challenges faced by stakeholders in the industry.

In response to these challenges, the emergence of digital technologies has paved the way for innovative solutions aimed at revolutionizing the real estate landscape. Online platforms have begun to democratize property transactions, offering greater accessibility, transparency, and convenience to buyers and sellers alike. However, existing platforms often fall short in providing a holistic solution that addresses the myriad complexities inherent in property transactions and management.

Against this backdrop, the PROPLED (Property Ledger) project emerges as a beacon of change in the real estate industry. Born out of the need for a comprehensive, user-centric platform that transcends the limitations of existing solutions, PROPLED is designed to redefine the way properties are bought, sold, and managed in the digital age.

PROPLED seeks to address the key pain points of the real estate industry by providing a centralized platform where users can access a wide range of functionalities, including property listing, tax disclosure, legal dispute resolution, and transparent ownership history. By leveraging cutting-edge technologies such as web development frameworks, secure authentication mechanisms, and robust data management systems, PROPLED aims to streamline the property transaction process while enhancing transparency, security, and efficiency.

Moreover, PROPLED is not just a transactional platform; it is a catalyst for empowerment and innovation in the real estate ecosystem. By empowering users with greater visibility and control over their property transactions, PROPLED aims to foster trust, collaboration, and mutual benefit among all stakeholders involved.

1.3 SCOPE OF PROPLED

The PROPLED (Property Ledger) project is ambitiously designed to revolutionize the real estate industry by offering a comprehensive online platform that facilitates property transactions and management. At its core, PROPLED aims to provide users with a seamless

experience from listing properties for sale to finalizing transactions, all within a secure and user-friendly environment. With a focus on transparency, efficiency, and user empowerment, the project endeavors to address the longstanding challenges faced by buyers, sellers, and administrators in the realm of property management.

The scope of this project is to develop a transparent and equitable property booking system that allows for the allocation of land properties to users. It includes features for multiple partners to be associated with one property, verification of tax and old owners' details, user-friendly interface, communication tools, and efficient data management systems. The project ensures the protection of user data and efficient management of bookings and property listings.

Central to the scope of the PROPLED project is the implementation of robust user authentication and authorization mechanisms. This ensures a secure environment where admins have the authority to approve or reject user registrations, while registered users are granted access to various features based on their roles and permissions. With security as a top priority, PROPLED aims to instill confidence among users and foster trust in the platform.

Transaction facilitation and dispute resolution are also integral components of the PROPLED project scope. A booking and payment system will streamline the reservation process for properties, while a messaging system will facilitate communication between buyers and sellers, enabling them to negotiate terms and finalize transactions. Admins will have access to a centralized dashboard where they can manage user registrations, property listings, reported issues, and disputes, ensuring effective oversight and administration of the platform. With a focus on security, privacy, and user-centric design, PROPLED aims to redefine the way properties are bought, sold, and managed in the digital age, ushering in a new era of efficiency and transparency in the real estate industry.

1.4 OBJECTIVES

The proposed system aims to revolutionize the property booking process by introducing transparency and efficiency into the allocation of land properties. Through the incorporation of innovative features such as multiple partner association and detailed verification procedures, users are empowered to make informed decisions, leading to a fair and equitable allocation process.

- To ensure transparency and accuracy in the allocation of land properties.
- To allow multiple partners to be associated with one property.
- To verify tax and old owners' details for each property.
- To provide users with all relevant information before making a booking.
- To avoid confusion and ensure fairness in the allocation process.

CHAPTER 2

LITERATURE SURVEY

2.1 EXISTING SYSTEM

The current system in real estate management relies heavily on manual paperwork, decentralized databases, and traditional record-keeping methods. These methods often result in inefficiencies, delayed processes, and errors due to the sheer volume of paperwork and the need for manual data entry and verification. Property data is siloed across multiple platforms, making it challenging to obtain a comprehensive view of property details. Accessibility to this information is often limited to specific stakeholders, with the general public or smaller investors having restricted access to critical data such as property ownership records, land boundaries, or historical transactions.

Moreover, data security concerns prevail, with vulnerabilities in the existing systems leading to potential data breaches, unauthorized access, and a lack of robust mechanisms to ensure data integrity. This situation poses significant challenges in maintaining the accuracy, confidentiality, and reliability of property information. Interoperability issues between different databases and systems hinder seamless data exchange and integration, resulting in discrepancies and duplication of efforts during data reconciliation processes.

Furthermore, the absence of standardized formats and protocols across various jurisdictions or agencies complicates data sharing and hampers efforts towards uniformity and consistency in property records. The lack of technological integration and the slow adoption of modern tools and platforms limit the ability to leverage advanced technologies for data analysis, predictive insights, and decision-making, thereby restraining the industry from harnessing the full potential of innovation in the real estate sector. These challenges collectively underscore the necessity for a more streamlined, integrated, and technologically advanced solution to address the shortcomings prevalent in the current real estate and land administration systems.

2.2 DRAWBACKS OF EXISTING SYSTEM

The drawbacks of the existing system are:

Fragmented Data Repositories: Existing systems often suffer from fragmented and siloed data repositories scattered across multiple sources. This fragmentation leads to inconsistencies, redundancies, and difficulties in accessing a comprehensive and unified view of property information.

Manual Processes and Paperwork: Traditional record-keeping methods heavily rely on manual processes and paperwork, making data entry, verification, and retrieval time-consuming and error-prone. The reliance on manual handling limits efficiency and increases the risk of inaccuracies.

Limited Accessibility and Transparency: Accessibility to property information is often restricted, with the general public or smaller stakeholders having limited access to crucial data like property ownership records, transaction history, or legal encumbrances. This lack of transparency can hinder trust and transparency within the industry.

Security Vulnerabilities: The existing systems are susceptible to security vulnerabilities, leading to concerns regarding data breaches, unauthorized access, and data tampering. Inadequate security measures compromise the integrity and confidentiality of property-related data.

Interoperability Challenges: Interoperability issues between disparate systems hinder seamless data exchange and integration. Incompatible formats and protocols often lead to data inconsistencies and duplication of efforts during data reconciliation processes.

Lack of Standardization: The absence of standardized formats and protocols across different jurisdictions or agencies creates challenges in data sharing and compromises uniformity and consistency in property records.

Limited Technological Integration: The slow adoption of modern technologies such as AI, blockchain, and advanced analytics restricts the potential for leveraging these innovations to enhance decision-making, data analysis, and user experiences within the real estate sector.

Inefficiencies and Delays: Overall, the existing systems' inefficiencies, reliance on manual processes, and lack of technological advancement result in delays, errors, and increased operational costs, inhibiting the sector from realizing its full potential.

2.3 PROPOSED SYSTEM

The proposed system aims to streamline the property booking process by offering a transparent and accurate method for allocating land properties to users. It allows multiple partners to be associated with one property, and the admin can upload and authenticate tax details and old owners' information to ensure that the allocation process is fair and legitimate. Users can view all relevant information, including the old owner's details, partners' information, and tax details, before making a booking. This feature helps to avoid any

confusion or misunderstandings during the booking process, and ensures that the allocation of land properties is done in a fair and transparent manner.

Centralized Data Repository: At the core of the proposed system is a centralized data repository that consolidates property-related information from diverse sources. This repository integrates data points such as survey numbers, ownership details, geographical specifics, legal records, transaction histories, and property valuations into a unified platform. The centralization of data ensures coherence, accuracy, and real-time updates, eliminating the inconsistencies and redundancies prevalent in fragmented systems.

User-Friendly Interface and Accessibility: The "PROPLED" platform offers an intuitive, user-friendly interface accessible to a wide range of stakeholders, including government entities, real estate agencies, property buyers, sellers, investors, and the general public. Through a well-designed dashboard and search functionalities, users can easily access comprehensive property details, including ownership history, encumbrances, land boundaries, valuations, and market trends. This enhanced accessibility fosters transparency and trust within the industry.

Advanced Technological Integration: The proposed system leverages cutting-edge technologies to enhance functionalities and provide value-added services. Features such as geospatial analytics enable mapping and visualization of property boundaries and land use patterns. Integration of blockchain technology ensures secure, tamper-proof record-keeping, enhancing data integrity and security. Additionally, AI-driven analytics offer predictive insights, aiding in informed decision-making for property transactions, investments, and legal procedures.

Standardization and Interoperability: "PROPLED" focuses on establishing standardized data formats, ensuring interoperability between different systems and jurisdictions. The platform adheres to industry standards and regulatory frameworks, enabling seamless data exchange and integration. By fostering uniformity and consistency in property records, it mitigates the challenges associated with disparate systems and data sources prevalent in the current landscape.

Enhanced Decision-Making and Industry Collaboration: The platform empowers stakeholders with comprehensive property insights, aiding in informed decision-making. Real-time access to accurate data facilitates quicker and well-informed property transactions, mitigating risks and uncertainties. Collaboration with

government bodies, industry associations, and real estate agencies drives widespread adoption, setting benchmarks for industry standards and best practices.

Comprehensive Property Information Management: The "PROPLED: Property Ledger" platform aims to revolutionize property information management by centralizing diverse data points related to real estate properties. It aggregates information such as survey numbers, land parcel details, ownership records, property valuations, legal encumbrances, historical transaction data, and zoning information. This comprehensive repository offers a holistic view of property assets, enabling users to access a complete and accurate portrayal of each property's attributes.

Intuitive User Interface and Accessibility: The platform features an intuitive and user-friendly interface designed to cater to a broad spectrum of users. The interface facilitates easy navigation and search functionalities, allowing stakeholders to retrieve specific property information swiftly. By providing a user-centric experience, the platform ensures seamless accessibility to property details for various stakeholders, including government agencies, property developers, legal professionals, buyers, and sellers.

2.3.1 ADVANTAGES OF THE PROPOSED SYSTEM

The proposed system is designed to offer a wide range of advantages. Here are the key advantages of the proposed system:

Enhanced Transparency and Trust: The "PROPLED" system fosters transparency by providing stakeholders with access to accurate and up-to-date property information. This transparency builds trust among buyers, sellers, investors, and regulatory authorities, fostering a more reliable and credible real estate ecosystem. Users can verify property details, ownership records, and historical transactions, thereby reducing potential disputes and enhancing trust in property dealings.

Collaborative Ecosystem and Industry Impact: "PROPLED" encourages collaboration among government bodies, real estate associations, regulatory authorities, and industry stakeholders. This collaborative ecosystem fosters knowledge sharing, sets industry benchmarks, and promotes the adoption of best practices and innovative technologies in property management. The platform's industry impact extends to driving efficiency, transparency, and trust within the real estate sector.

Streamlined Administrative Processes: By centralizing property-related data and automating administrative processes, "PROPLED" streamlines workflows for government entities and regulatory bodies. The system facilitates smoother property

registrations, title transfers, and regulatory compliance procedures. Automation reduces paperwork, accelerates approvals, and minimizes bureaucratic hurdles, enhancing overall operational efficiency.

Cost Savings and Operational Efficiency: The adoption of "PROPLED" leads to cost savings and increased operational efficiency. By minimizing manual data entry, reducing paperwork, and optimizing processes, the system decreases operational costs associated with data management, storage, and maintenance. Efficiency gains result from quicker access to information, reducing time spent on research and verification.

Standardization and Interoperability: "PROPLED" emphasizes data standardization and interoperability, adhering to industry standards and protocols. This standardization fosters seamless data exchange and integration across different systems, ensuring uniformity and consistency in property records. By promoting interoperability, the platform mitigates data inconsistencies, duplication of efforts, and enhances data reliability, facilitating collaboration among stakeholders.

CHAPTER 3

SYSTEM DESIGN

3.1 DEVELOPMENT ENVIRONMENT

The purpose of the Development Environment is to produce the specification of the analysis task and to establish complete information about the requirement, behavior and also the other constraints like functional performance and so on. The main aim of the Requirement Specification is to completely specify the technical requirements for the software product in a concise and unambiguous manner.

HARDWARE REQUIREMENTS

The hardware requirements may serve as the basis for a contract for the implementation of the system and should therefore be a complete and consistent specification of the whole system. They are used by software engineers as the starting point for the system design. It shows what the systems do and not how it should be implemented.

- Hard disk: Intel Pentium
- Ram: 4GB
- Hard disk: 260GB

SOFTWARE REQUIREMENTS

The software requirements are the specification of the system. It should include both a definition and a specification of requirements. It is a set of what the system should do rather than how it should do it. The software requirements provide a basis for creating the software requirements specification. It is useful in estimating cost, planning team activities, performing tasks, and tracking the team's and tracking the team's progress throughout the development activity.

- Operating System: Windows 7 or higher
- Language: Java, JSP
- Database: MySQL
- Tool: NetBeans 7.3.1

JAVA

The software component of this project is developed using Java, a versatile and widely-used programming language renowned for its platform independence and robustness. Leveraging Java's object-oriented paradigm and extensive libraries, the system offers a scalable and efficient solution for property booking and management. Java's flexibility enables seamless

integration with various technologies, facilitating the implementation of features such as user authentication, property listing, communication tools, and data management systems. With its reliability, performance, and extensive community support, Java serves as the foundation for building a feature-rich and user-friendly platform that streamlines the property booking process while ensuring transparency and accuracy in the allocation of land properties.

JSP

The software component of this project utilizes JSP (JavaServer Pages) to develop dynamic and interactive web pages for the property booking system. JSP provides a convenient and efficient way to separate the presentation layer from the business logic, allowing for the creation of visually appealing and responsive user interfaces. Leveraging JSP's tag-based syntax and seamless integration with Java code, the system delivers a seamless user experience, enabling users to navigate property listings, communicate with partners, and manage bookings effortlessly. With its ability to generate dynamic content and interact with server-side Java components, JSP plays a pivotal role in realizing the project's objectives of simplifying the property booking process while ensuring transparency and accuracy in property allocation.

NETBEANS

The software component of this project is developed using NetBeans, an integrated development environment (IDE) that provides comprehensive tools and features for Java development. NetBeans offers a user-friendly interface and powerful capabilities for coding, debugging, and deploying Java applications, making it an ideal choice for building the property booking system. Leveraging NetBeans' intuitive interface, code assistance, and project management tools, developers can efficiently create, test, and deploy the system's components, including user authentication, property listing, communication features, and data management systems. With its robust features and seamless integration with Java technologies, NetBeans accelerates the development process and ensures the delivery of a high-quality, reliable, and scalable solution for simplifying property transactions while ensuring transparency and accuracy in property allocation, as in Fig 3.1.

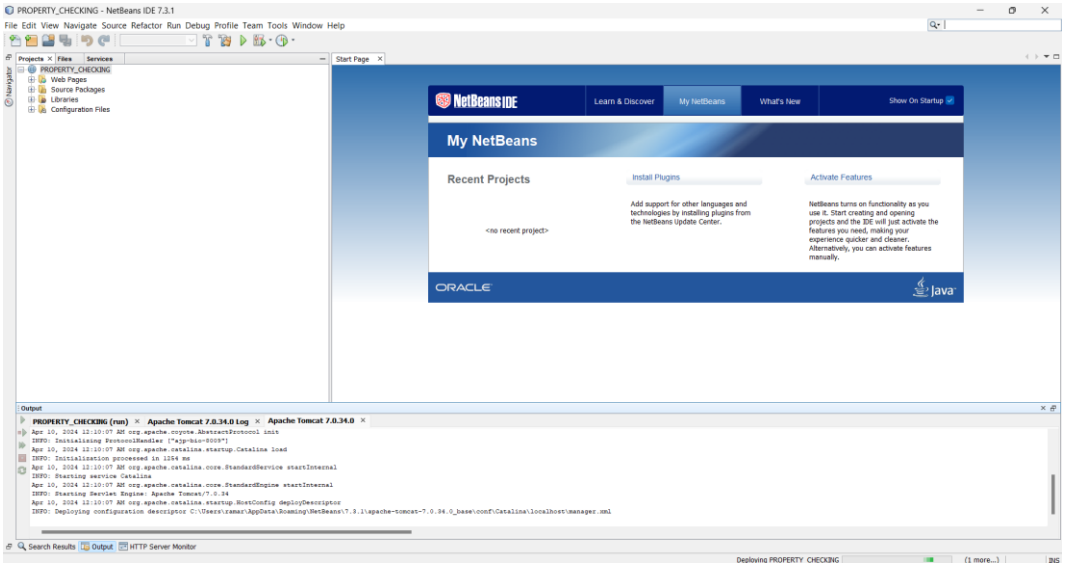


Fig 3.1. NetBeans Interface

MYSQL

The software component of this project relies on MySQL, a powerful relational database management system, to store and manage the vast array of property-related data. MySQL provides a robust and scalable solution for organizing property listings, user information, booking records, and other essential data points. Leveraging MySQL's speed, reliability, and ACID compliance, the system ensures efficient data retrieval and manipulation, facilitating seamless property transactions and management. With its support for complex queries, indexing, and transactional integrity, MySQL serves as the backbone of the project, enabling the realization of objectives such as transparency, accuracy, and user-friendly data management in the property booking process, as in Fig 3.2.

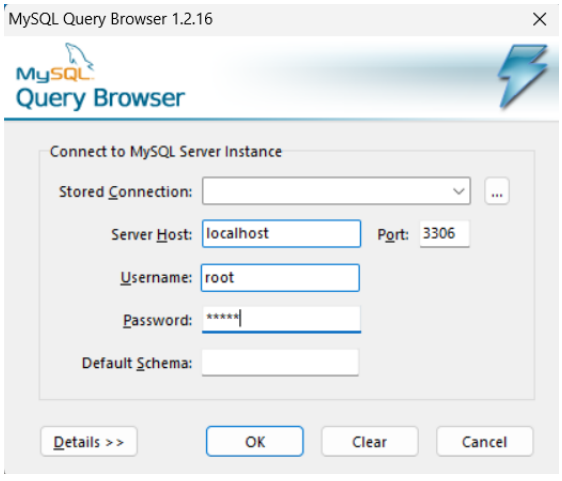


Figure 3.2. MySQL Query Browser

3.2 SYSTEM ARCHITECTURE DIAGRAM

The system architecture diagram, shown in Fig 3.3, for the "PROPLED: Property Ledger" project serves as a visual representation depicting the arrangement, interaction, and dependencies among various system components. This diagram is crucial for conveying the high-level structure and design of the entire system in a clear and understandable manner.

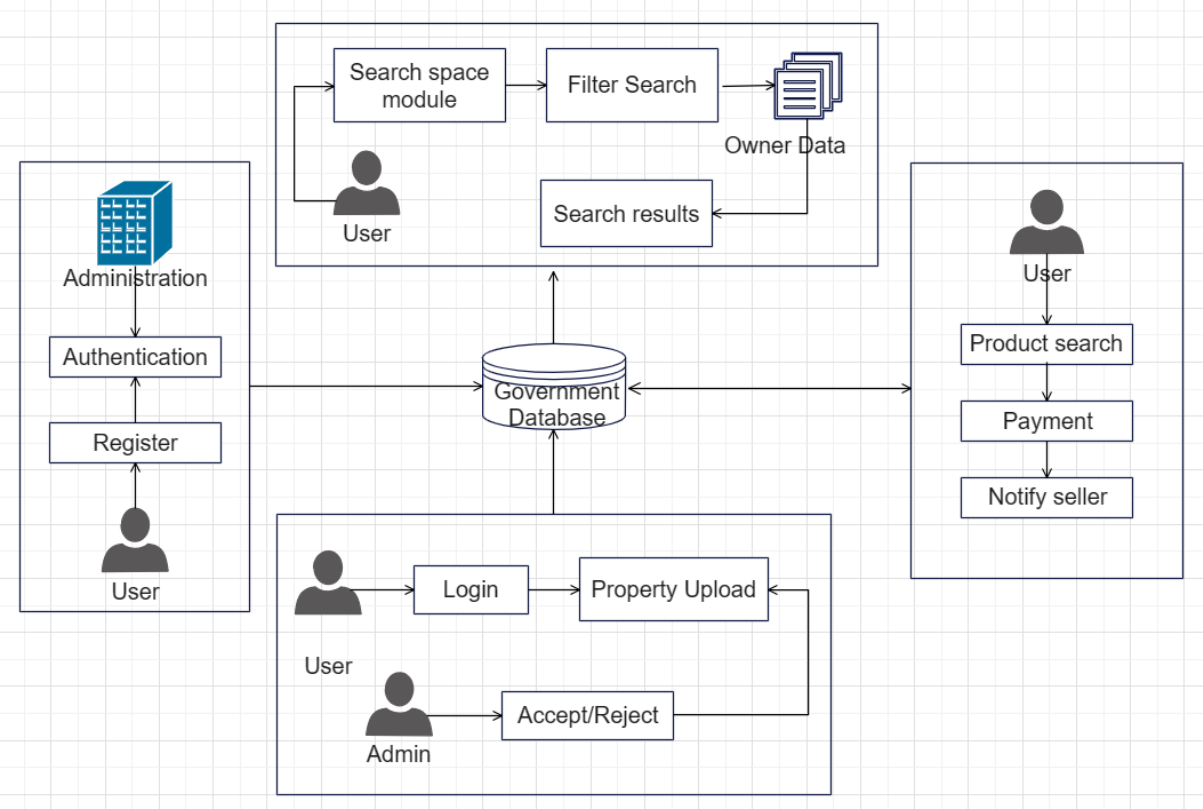


Fig 3.3. Architecture Diagram of PROPLED

The architectural diagram has four modules. The modules are login module, User authentication module, Help module and Search space module. User interacts and perform their function with each of the modules. In the login module the users will login themselves into propled to carry on with further search. In the User authentication module each and every user is authenticated based on their KYC details to ensure integrity. In the help module users can search out their queries regarding this system and clarify it using the provided documentation. In the Search space module users can filter the desired location by providing the appropriate property details to acquire the property owner’s details.

3.3 DESIGN OF THE ENTIRE SYSTEM

USE CASE DIAGRAM

The software design of this project incorporates a comprehensive use case diagram to illustrate the various interactions and functionalities of the property booking system. The use case diagram visually represents the different actors, such as admins and users and their corresponding roles and interactions with the system. Fig 3.4 outlines primary use cases such as user registration, property listing, booking management, communication between users, and administration tasks. By visually depicting the system's functionalities and user interactions, the use case diagram serves as a blueprint for the development team, guiding the implementation process and ensuring alignment with the project's objectives of simplifying property transactions, ensuring transparency, and facilitating efficient property management.

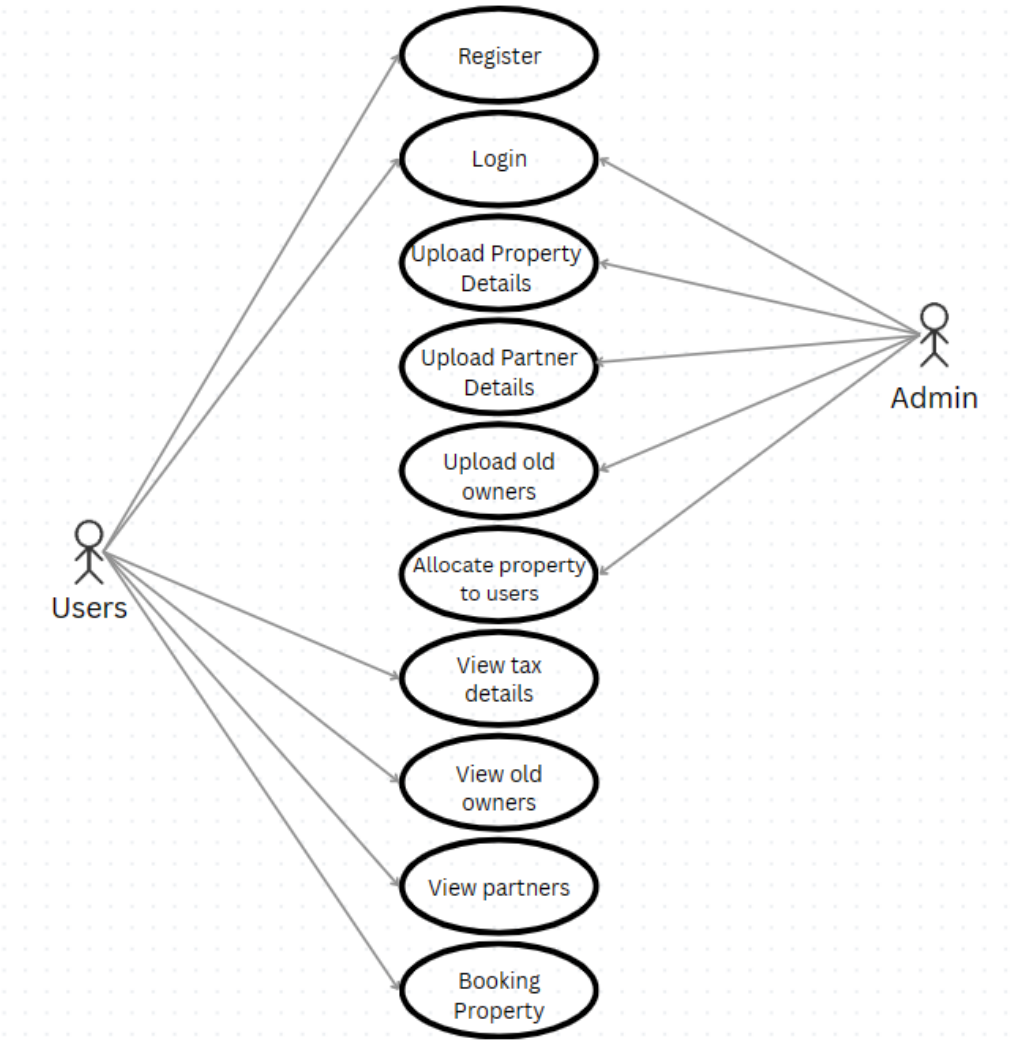


Fig 3.4. Use Case Diagram of the Propled

SEQUENCE DIAGRAM

The software design of this project includes a sequence diagram that illustrates the sequence of interactions between different components and actors within the property booking system. The sequence diagram outlines the chronological flow of messages exchanged between actors such as users, admins, and the system itself during various scenarios such as user registration, property listing, booking confirmation, and communication between users. By visually representing the sequence of actions and messages, the sequence diagram provides valuable insights into the system's behavior and helps ensure that the interactions between different components are well-coordinated and in accordance with the project's objectives of simplifying property transactions, ensuring transparency, and facilitating efficient property management.

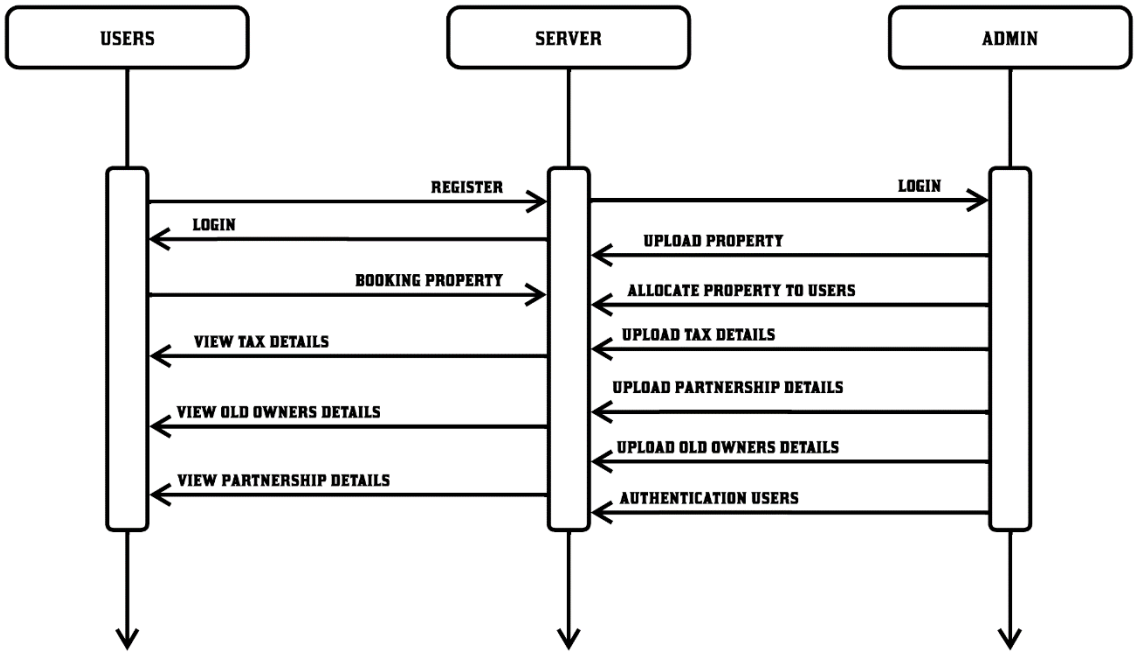


Fig 3.5. Sequence Diagram of the Propled

CHAPTER 4

SYSTEM DESCRIPTION

4.1 DATA COLLECTION

The data collection process of the PROPLED (Property Ledger) project involves gathering and organizing various types of information related to property transactions and management. This includes but is not limited to property details, user information, tax dues, legal disputes, ownership history, booking records, and communication logs.

Property Details - Information such as property type, location, size, amenities, photos, pricing details, and availability status are collected from users who wish to list their properties for sale.

User Information - User registration data, including personal details and authentication credentials, is collected during the registration process. Admins verify and approve user registrations to ensure a secure environment.

Tax Dues and Legal Disputes - Users are required to disclose any tax dues or legal disputes associated with their properties. This information is crucial for ensuring transparency and facilitating informed decision-making by potential buyers.

Ownership History - A transparent record of ownership history for each property is maintained, showcasing previous owners and any changes in ownership over time. This data adds credibility to property listings and enhances trust among users.

Booking Records - Details of property bookings, including booking costs, payment transactions, and booking statuses, are collected to facilitate the reservation process and track property availability.

The data collection process adheres to privacy regulations and security protocols to ensure the confidentiality and integrity of user information. By systematically gathering and organizing relevant data points, the PROPLED project aims to create a comprehensive platform that empowers users in their property transactions while promoting transparency, fairness, and efficiency.

4.2 OBJECT CONSTRUCTION

The object construction phase of the PROPLED (Property Ledger) project involves the creation and instantiation of various objects that represent key entities and functionalities within the system. These objects encapsulate data and behavior, enabling the system to manage property transactions and user interactions effectively.

Property Object: The Property object represents individual properties listed on the platform. It encapsulates attributes such as property type, location, size, amenities, pricing details, and availability status. Methods associated with the Property object facilitate actions such as listing properties for sale, updating property information, and managing bookings.

User Object: The User object represents registered users of the platform, including admins, property owners, and potential buyers. It stores user details such as username, password, email, and role. The User object provides methods for user authentication, registration, profile management, and communication with other users.

Booking Object: The Booking object manages property bookings initiated by users. It stores booking details such as booking ID, property ID, booking status, booking cost, and payment information. Methods associated with the Booking object handle booking requests, payment processing, and booking status updates.

Admin Object: The admin object represents administrators responsible for managing the platform. It stores admin credentials and privileges, enabling admins to perform actions such as user authentication, property approval, dispute resolution, and system maintenance.

4.3 ALGORITHM

The system implementation of the PROPLED (Property Ledger) project involves the development and implementation of algorithms to perform various tasks efficiently and accurately. These algorithms play a crucial role in the functionality and performance of the system, enabling tasks such as property listing, user authentication, booking processing, and communication management to be executed seamlessly.

- The user authentication algorithm verifies the identity of users attempting to access the system. It compares the provided credentials (username and password) against stored user data in the database. If the credentials match, the user is granted access to the system; otherwise, access is denied.
- The property listing algorithm facilitates the posting of properties for sale by users. It collects property details provided by the user through the system interface and stores them in the database. The algorithm validates the input data to ensure accuracy and completeness before adding the property listing to the platform.
- The booking processing algorithm manages property bookings initiated by users. It validates booking requests, checks property availability, calculates booking costs, and

handles payment transactions. The algorithm updates the booking status and notifies users of the booking confirmation or rejection.

4.4 METHODOLOGY

The system implementation of the PROPLED (Property Ledger) project follows a structured and iterative methodology to ensure the successful development, testing, and deployment of the software system. The methodology encompasses several key stages, each of which plays a crucial role in the overall development process.

- The methodology begins with a thorough analysis of the project requirements, including stakeholder needs, system functionalities, and technical specifications. This stage involves gathering input from stakeholders, defining system objectives, and documenting functional and non-functional requirements to serve as the foundation for system development.
- Once the requirements are clearly defined, the design phase commences, where system architecture, data models, and user interfaces are conceptualized and documented. This phase involves creating detailed design specifications that outline the structure and behavior of the system components, ensuring alignment with the project objectives and requirements.
- With the design specifications in place, the implementation phase begins, where the actual coding and development of the software system take place. Skilled developers utilize programming languages, frameworks, and tools such as Java, HTML, CSS, JavaScript, and NetBeans to translate the design specifications into functional software components. The implementation phase follows best practices, coding standards, and version control procedures to ensure code quality, readability, and maintainability.
- Rigorous testing procedures are conducted throughout the implementation phase to verify the functionality, reliability, and performance of the software system. This includes unit testing to validate individual components, integration testing to assess the interoperability of different modules, and system testing to evaluate the system as a whole. Testing is conducted using a combination of manual and automated techniques to identify and rectify any defects or inconsistencies.

Upon successful testing, the software system is deployed to a production environment where it is accessible to users. Deployment involves configuring servers, setting up databases, and deploying application code to web servers or cloud platforms. Continuous monitoring, maintenance, and support are essential post-deployment to ensure optimal performance, security, and availability of the system.

CHAPTER 5

RESULTS AND DISCUSSION

SCREENSHOTS

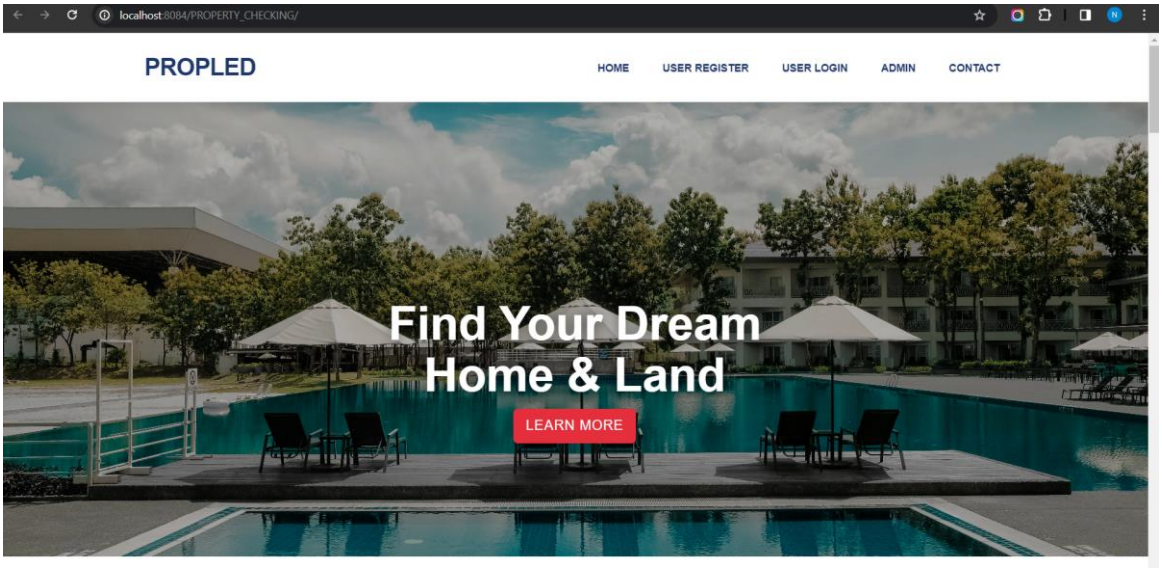


Fig 5.1. Welcome Screen

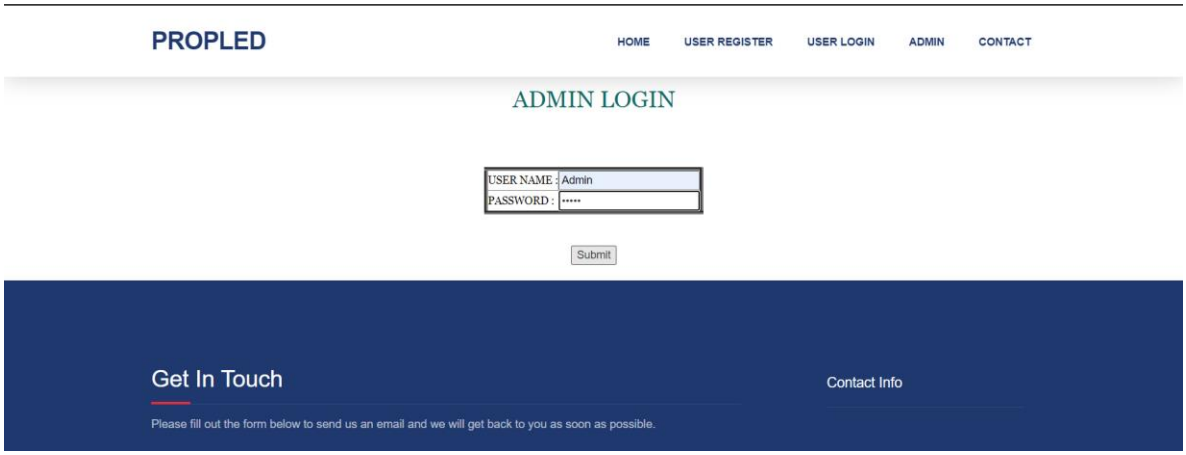


Fig 5.2. Admin Login Page

This is the login page where the users and admins can login and perform to perform their desired actions. If a new user wants to register themselves, they can do the same by providing the users required details, which will be further sent to admins for approval.

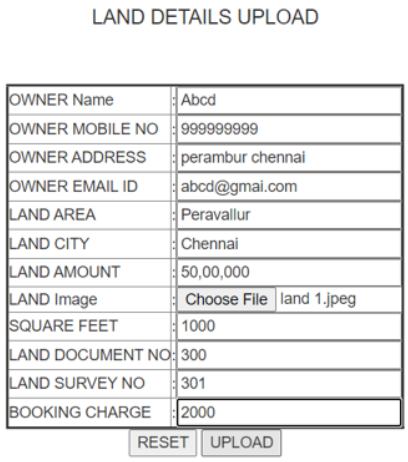


Fig 5.3. Upload Property Details

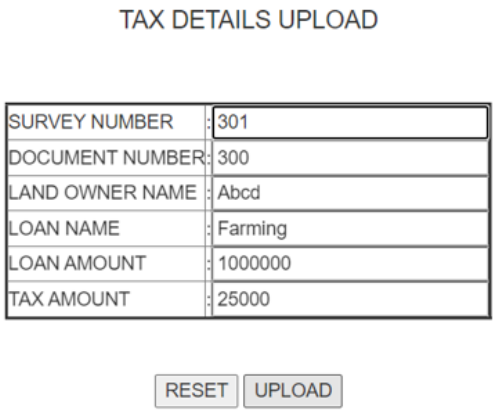


Fig 5.4. Tax Details Upload Page

Once the admin logins he/she will be directed to upload the property details like owner name, mobile number, email address, land area and amount after which they must upload the tax details like name, amount, pending tax and other required details. The user interface of the PROPLED platform has been meticulously designed to offer an intuitive and visually appealing experience. Users can effortlessly navigate through property listings, submit booking requests, communicate with other users, and manage their accounts. This seamless user experience enhances engagement and satisfaction among users, contributing to the overall success of the platform.

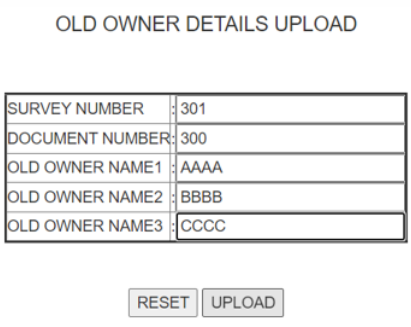


Fig 5.5. Old Owners' Details Update

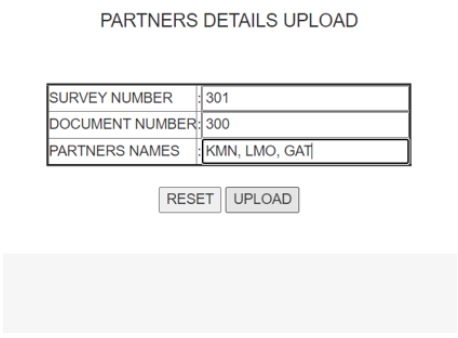


Fig 5.6. Partners Details Upload

Old owners' details of the property are updated by the admin in this page, the details of last three owners of the property are collected to make the current buyer know about the property's history. Then the partnership details are provided by the admin like name of one or more partners for the buyer to know from whom to buy the property.

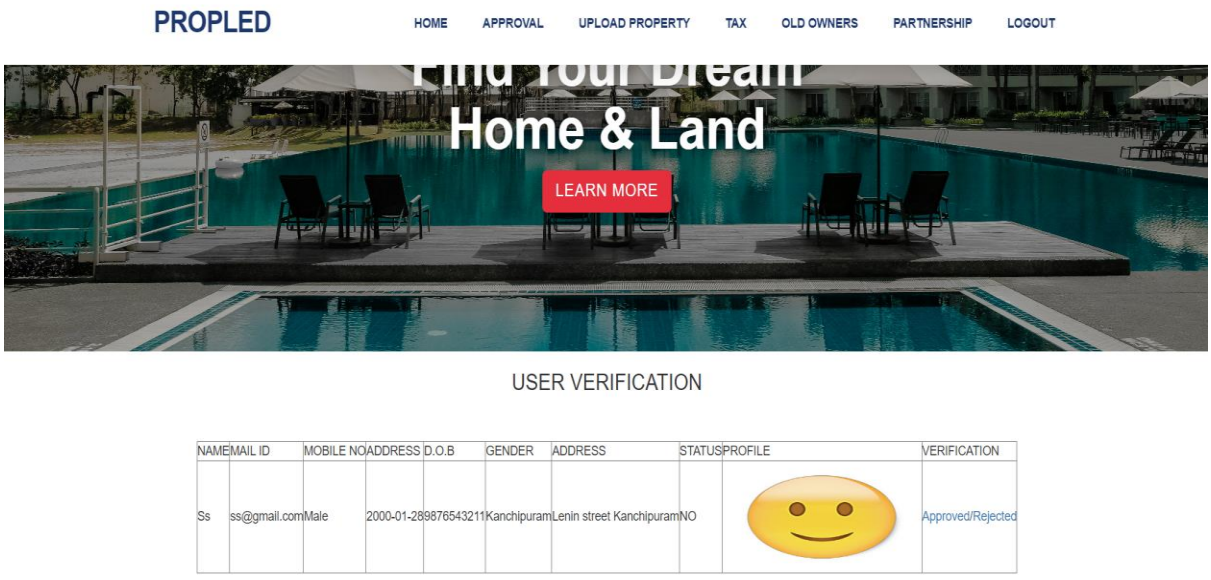


Fig 5.7. Pending Approval Screen

This pending approval screen represents the approvals that are yet to be approved or rejected by the admins, these requests are sent from new users to upload their property.



Fig 5.8. New User Registration



Fig 5.9. User Login

Existing users are allowed to login whereas new users are supposed to register themselves in this page only after the approval from the admins, authenticated users are allowed to post and buy properties.

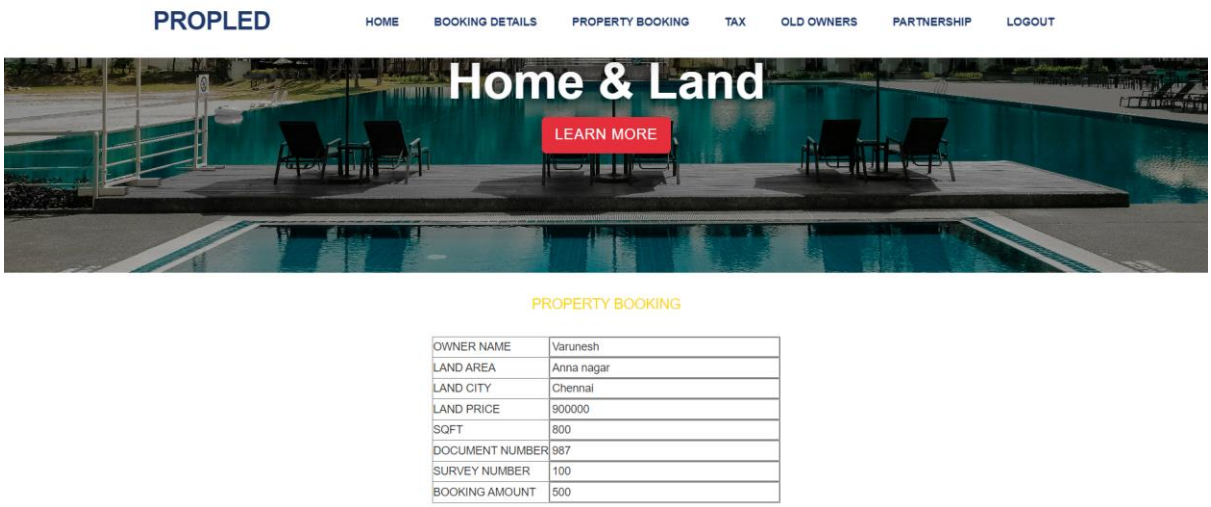


Fig 5.10. Booking Details

When users pay certain amount and book a property it will be listed in the booking details tab next to home page.



Fig 5.11. Property Booking

Property booking page allows the users to view available properties for sale and the specified property's details. Each and every property is unique, land area and other details of the properties are listed for user's reference. If they like the property they can book the same by paying the mentioned advance amount.

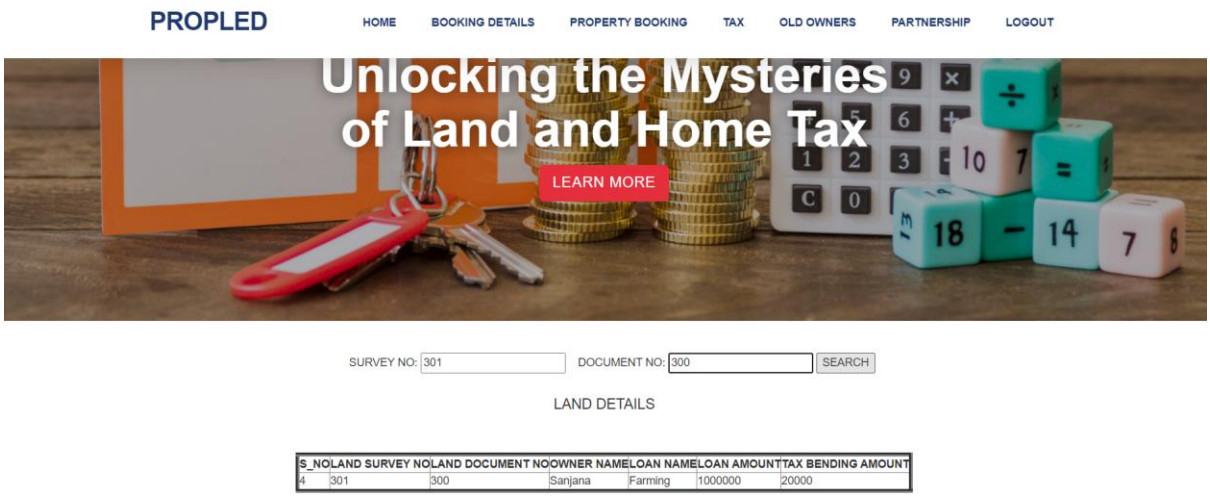


Fig 5.12. Tax Pending Details

In the Tax tab search space for survey number and document number are available where users can enter a specific property's survey number and document number to get the tax details.

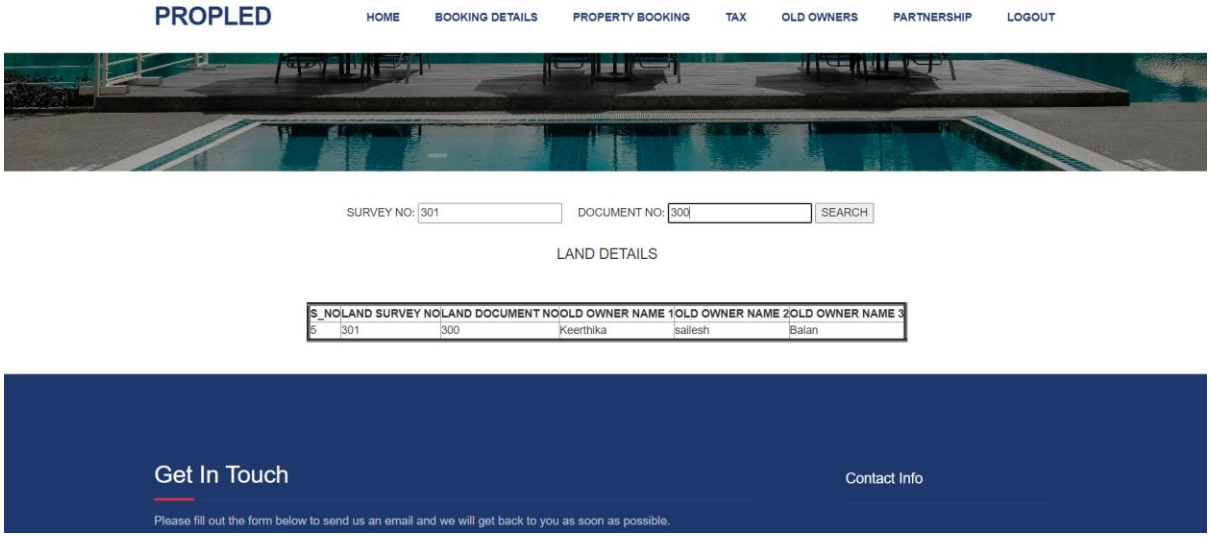


Fig 5.13. Old Owners Details

In the old owners tab search space for survey number and document number are available where users can enter a specific property's survey number and document number to get the old owners details. Users can trust the PROPLED platform to uphold the confidentiality, integrity, and availability of these data since its only updated by authenticated users and admins.

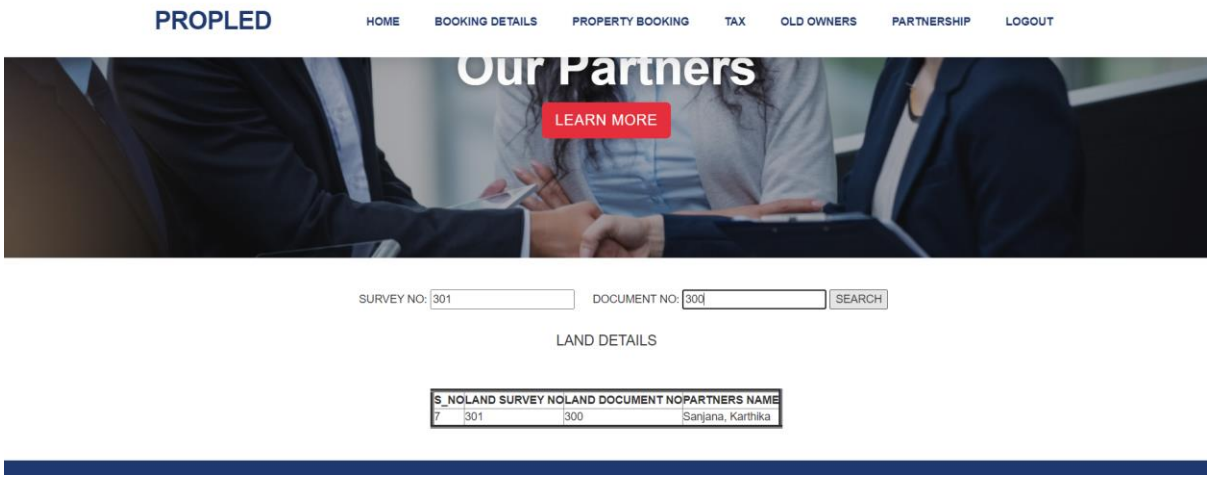


Fig 5.14. Partners Details Display

In the Partnership tab search space for survey number and document number are available where users can enter a specific property's survey number and document number to get the details of current partners of the property. The successful development of the PROPLED platform, offers a comprehensive solution for property transactions and management. Its user-friendly interface, robust functionality, high performance, and stringent security measures makes it a valuable asset for property owners, buyers, and administrators alike. Through continuous feedback, monitoring, and improvement efforts, the PROPLED system remains at the forefront of innovation and excellence in the real estate industry.

CHAPTER 6

BUSINESS CANVAS MODEL

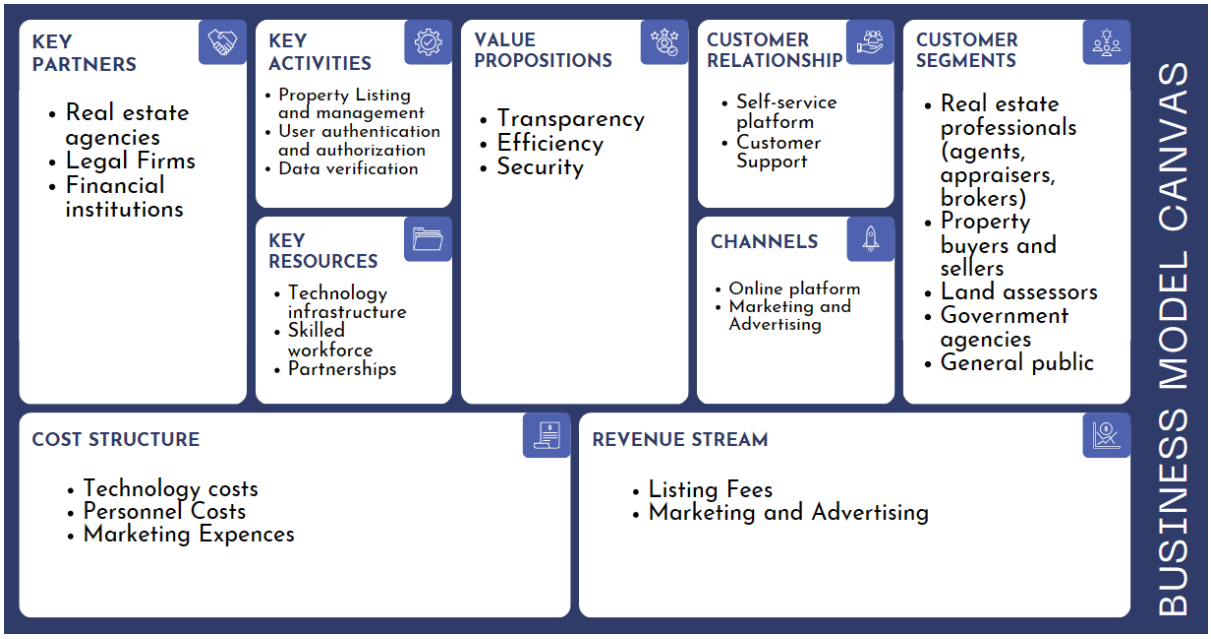


Fig 6.1. Business Canvas Model

Key Partnerships

- Real Estate Agencies: Collaborate with established real estate agencies to access a wider network of property listings and potential buyers.
- Legal Firms: Partner with legal firms to ensure compliance with regulatory requirements and facilitate legal documentation for property transactions.
- Financial Institutions: Forge partnerships with banks or financial institutions to offer financing options for property purchases.

Key Activities

- Property Listing and Management: Develop and maintain a digital platform for property owners to list their properties, manage bookings, and communicate with potential buyers.
- User Authentication and Authorization: Implement robust authentication mechanisms to verify user identities and ensure the security of transactions.
- Data Verification: Verify property details, tax dues, ownership history, and legal documentation to provide accurate and reliable information to users.

Key Resources

- Technology Infrastructure: Invest in robust servers, databases, and security systems to support the platform's operations and ensure data integrity.
- Skilled Workforce: Hire skilled developers, designers, and support staff to develop, maintain, and support the platform.
- Partnerships: Leverage partnerships with real estate agencies, legal firms, and financial institutions to access resources and expertise in the industry.

Value Proposition

- Transparency: Provide transparent information about property listings, tax dues, legal documentation, and ownership history to facilitate informed decision-making by users.
- Efficiency: Streamline property transactions and management processes, saving time and effort for property owners, buyers, and administrators.
- Security: Ensure the security and confidentiality of user data and transactions through robust authentication mechanisms and encryption protocols.

Customer Relationships

- Self-Service Platform: Offer a user-friendly platform where users can independently list properties, search for properties, initiate bookings, and manage transactions.
- Customer Support: Provide customer support services through various channels such as email, chat, and phone to assist users with any queries or issues they may encounter.

Channels

- Online Platform: Utilize a digital platform accessible via web browsers and mobile devices to reach a wide audience of property owners, buyers, and administrators.
- Marketing and Advertising: Implement digital marketing strategies such as social media advertising, search engine optimization (SEO), and content marketing to attract users to the platform.

Customer Segments

- Real Estate Professionals: Real estate agents, brokers, and developers use PROPLED to showcase properties and facilitate transactions efficiently.
- Property Buyers and Sellers: Homebuyers, investors, and property sellers rely on PROPLED for accurate property listings and transaction management.
- Land Assessors: Appraisers and surveyors access PROPLED for reliable property data and documentation to assist in valuation and surveying activities.

- Government Entities: Regulatory bodies and urban planning departments utilize PROPLED for transparent access to property information and regulatory oversight.
- General Public: Community members access PROPLED for researching property values, exploring real estate trends, and staying informed about local developments.

Revenue Streams

- Listing Fees: Charge property owners a fee for listing their properties on the platform, based on factors such as property type, location, and duration of listing.
- Transaction Fees: Earn revenue through transaction fees charged for property bookings and sales facilitated through the platform.
- Premium Features: Offer premium features or subscription plans for advanced functionalities such as enhanced property visibility, priority support, and analytics.

Cost Structure

- Technology Costs: Allocate resources for infrastructure maintenance, software development, and technology upgrades to support the platform's operations.
- Personnel Costs: Budget for salaries, benefits, and training for employees involved in development, maintenance, and customer support roles.
- Marketing Expenses: Allocate funds for marketing and advertising campaigns to promote the platform and attract users.

CHAPTER 7

CONCLUSION

CONCLUSION

In conclusion, the Property Ledger (PROPLED) has the potential to revolutionize the way land properties are allocated to users. By enhancing transparency, streamlining the process, and providing users with the necessary information to make informed decisions, the system can help to address the challenges associated with the existing system. The inclusion of efficient data management tools will ensure the protection of user data and facilitate the management of bookings and property listings. The proposed system has the potential to create a fair and equitable allocation process that enhances the user experience and builds trust in the property booking process.

FUTURE ENHANCEMENT

Future enhancements for the PROPLED platform could include the integration of advanced technologies such as augmented reality (AR) for immersive property viewing experiences, predictive analytics for providing insights on property trends and investment opportunities, and social integration features to foster collaboration and knowledge-sharing among users. Furthermore, customization options for map views, property data filters, and notification preferences could be implemented to offer users a personalized and tailored experience. Accessibility features will also be prioritized to ensure that the platform is inclusive and accessible to users with disabilities. By continuously innovating and integrating cutting-edge technologies, PROPLED aims to remain at the forefront of the real estate industry, providing users with an intuitive, transparent, and engaging platform for property transactions and management.

REFERENCES

1. Ashish Gupta; Inderpal Singh Mumick, “The Dynamic Homefinder: Evaluating Dynamic Queries In A Real-Estate Information Exploration System” Oct 1999. (Vol-4 Issue-2 IEEE Xplore, Online ISBN: 9780262287500)
2. Baoxuan Jin; Weiwei Song, “Cloud Infrastructure and Monitoring System for Real Estate Registration” June 2018. (IEEE Xplore DOI:10.1109/GEOINFORMATICS.2018.8557182)
3. Dhruv Goragandhi; Archana Nanade, “The Application of Blockchain as a Distributed Ledger and Smart Contract for Property Registration” Nov 2021 (IEEE Xplore, DOI: 10.1109/CENTCON52345.2021.9687997)
4. Duc Nha Le; Loc Le Tuan; Minh Nguyen Dang Tuan’ “Smart-Building Management System” IoT Application Business Model in Vietnam” Jan 2019 (Volume 141 TFSC).
5. Guo M.H., Liu Z.N., Mu T.J.,‘‘Beyond self-attention: External attention using two linear layers for visual tasks,’’ 2023, arXiv:2105.02358.
6. IBM, State Street Corp. Hyper ledger Fabric: A Distributed Operating System for Permissioned Innovative Resistance.
7. Jinyang Li John Jannotti, “A Scalable Location Service for Geographic Ad Hoc Routing”
8. Madakam S., & Kollu, S. Blockchain Technologies FundamentalsaPerceptions, Principles, Procedures and Practices. Journal of ˆ Social and Management Sciences, 34.
9. Michael Graglia .J, Christopher Mellon, Innovative Resistance and Property in 2018 : At end of the Beginning.
10. Miroslav Stefanovic, Dorde Przulj, Darko Stefanovic Blockchain and Land Administration: Possible applications and limitation. (2018)
11. Qianchuan Zhao; Ziyan Jiang, “Inspect Intelligent Building(I2B): A New Architecture of Building Control Systems Based on IoT” August 2019. (Volume 890 ISBN: = 978-981-13-6732-8) Citation: 2000. (Vol-3 Issue- 1 https://doi.org/10.1145/345910.345931)
12. Raquel Benbunan Fich, Arturo Castellanos, Digitalization of Land Records: From Paper to Innovative Resistance. (2018)
13. Siddhartha Sen, Sripati Mukhopadhya, “A Blockchain based Framework for Property Registration System in E-Governance” August 2021. (Vol-2 Issue-1 Research gate, DOI:10.5815/ijieeb.2021.04.03)

14. Tinu N.S., A Survey on Innovative Resistance Technology Taxonomy, Consensus Algorithms and Applications.
15. Tran Phuong Thao; Mhd Irvan, “Self-enhancing GPS-Based Authentication Using Corresponding Address” June 2020 (Research gate, 34th Annual IFIPWG)
16. Tsung Ting Kuo, Hugo Zavaleta Rojas and Lucila OhnoMachado(2019) Comparison of blockchain platforms: a systematic revie and healthcare examples.
17. Venus Brom Lima; Shima Ramesh Maniyath, “Geometric Location Finder Based on Encrypted Spatial Data Using Geometric Range Queries” April 2018. (IEEE Xplore, DOI: 10.1109/ICDI3C.2018.00024)
18. Xiaofang Yuan Wuhan, Ji-Hyun Lee, “User-oriented recommendation system for real estate websites” April 2023 (Volume -1 Issue -3 Research gate, DOI:10.1016/j.is.2012.08.004)
19. Yacov Manevich, Artem Barger, and Yoav Toc, Service Discovery for Hyperledger Fabric.
20. Yashwanth Madaka, Building a blockchain application using Hyperledger Fabric with Angular Frontend: Part-2 https://medium.com/coinmonks/building-a-blockchain-applicationusing-hyperledger-fabric-with-angular-frontend-part-2-22ef7c77f53.

APPENDIX

CODE

Index.jsp

```
<%@page contentType=”text/html” pageEncoding=”UTF-8”%>
<!DOCTYPE html>
<html>
   <head>
      <meta http-equiv=”Content-Type” content=”text/html; charset=UTF-8”>
      <title>JSP Page</title>
      <link rel=”stylesheet” href=”css/style.css”>
      <link rel=”stylesheet” type=”text/css”  href=”css/bootstrap.css”>
      <link rel=”stylesheet” type=”text/css” href=”fonts/font-awesome/css/font-
awesome.css”>
   </head>
   <body>
<nav id=”menu” class=”navbar navbar-default navbar-fixed-top”>
  <div class=”container”>
   <!—Brand and toggle get grouped for better mobile display →
   <div class=”navbar-header”>
     <button type=”button” class=”navbar-toggle collapsed” data-toggle=”collapse” data-
target=”#bs-example-navbar-collapse-1”> <span class=”sr-only”>Toggle navigation</span>
<span class=”icon-bar”></span> <span class=”icon-bar”></span> <span class=”icon-
bar”></span> </button>
     <a class=”navbar-brand page-scroll” href=”#page-top”>propled </a>
   </div>
    <!—Collect the nav links, forms, and other content for toggling →
   <div class=”collapse navbar-collapse” id=”bs-example-navbar-collapse-1”>
    <ul class=”nav navbar-nav navbar-right”>
      <li><a href=”index.jsp” class=”page-scroll active1”>Home</a></li>
      <li><a href=”USER_REGISTER.jsp” class=”page-scroll”>USER REGISTER</a></li>
      <li><a href=”USER_LOGIN.jsp” class=”page-scroll”>USER LOGIN</a></li>
      <li><a href=”ADMIN.jsp” class=”page-scroll”>ADMIN</a></li>
      <li><a href=”#contact” class=”page-scroll”>Contact</a></li>
    </ul>
   </div>
   <!-- /.navbar-collapse →
  </div>
</nav>
<!—Header →
<header id=”header”>
  <div class=”intro”>
   <div class=”overlay”>
    <div class=”container”>
      <div class=”row”>
        <div class=”col-md-8 col-md-offset-2 intro-text”>
          <h1>Find Your Dream <br>
           Home & Land</h1>
```

```
          <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis sed dapibus leo nec ornare diam. Sed commodo nibh ante facilisis 34ilename dolor feugiat at.</p>
          <a href=”#about” class=”btn btn-custom btn-lg page-scroll”>Learn More</a> </div>
        </div>
      </div>
    </div>
  </div>
</header>

<!—Services Section →
<div id=”services”>
  <div class=”container”>
    <div class=”section-title”>
      <h2>We Provide</h2>
    </div>
    <div class=”row”>
      <div class=”col-md-4”>
        <div class=”service-media”> <img src=”img/services/apartment.jpg” alt=” “> </div>
        <div class=”service-desc”>
          <h3>Apartment</h3>
          <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis sed dapibus leo nec ornare diam sedasd commodo nibh ante facilisis 34ilename dolor feugiat at.</p>
        </div>
      </div>
      <div class=”col-md-4”>
        <div class=”service-media”> <img src=”img/services/home.jpg” alt=” “> </div>
        <div class=”service-desc”>
          <h3>Home</h3>
          <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis sed dapibus leo nec ornare diam sedasd commodo nibh ante facilisis 34ilename dolor feugiat at. Lorem ipsum dolor sit amet, consectetur adipiscing elit.</p>
        </div>
      </div>
      <div class=”col-md-4”>
        <div class=”service-media”> <img src=”img/services/villa.jpg” alt=” “> </div>
        <div class=”service-desc”>
          <h3>Villa</h3>
          <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis sed dapibus leo nec ornare diam sedasd commodo nibh ante facilisis 34ilename dolor feugiat at.</p>
        </div>
      </div>
    </div>
    <div class=”row”>
      <div class=”col-md-4”>
        <div class=”service-media”> <img src=”img/services/property1.jpg” alt=” “> </div>
        <div class=”service-desc”>
          <h3>Property </h3>
          <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis sed dapibus leo nec ornare diam sedasd commodo nibh ante facilisis 34ilename dolor feugiat at.</p>
        </div>
```

```
    </div>
    <div class=”col-md-4”>
      <div class=”service-media”> <img src=”img/services/property6.jpg” alt=” “> </div>
      <div class=”service-desc”>
        <h3>Land</h3>
        <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis sed dapibus leo nec ornare diam sedasd commodo nibh ante facilisis 35ilename dolor feugiat at.</p>
      </div>
    </div>
    <div class=”col-md-4”>
      <div class=”service-media”> <img src=”img/services/building.jpg” alt=” “> </div>
      <div class=”service-desc”>
        <h3>Building</h3>
        <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis sed dapibus leo nec ornare diam sedasd commodo nibh ante facilisis 35ilename dolor feugiat at.</p>
      </div>
    </div>
  </div>
 </div>
</div>
<!—Gallery Section →
<div id=”portfolio”>
 <div class=”container”>
  <div class=”section-title”>
   <h2>Our Properties</h2>
  </div>
  <div class=”row”>
   <div class=”portfolio-items”>
    <div class=”col-sm-6 col-md-4 col-lg-4”>
     <div class=”portfolio-item”>
      <div class=”hover-bg”> <a href=”img/portfolio/01-large.jpg” title=”Project Title” data-lightbox-gallery=”gallery1”>
       <div class=”hover-text”>
        <h4>Lorem Ipsum</h4>
       </div>
       <img src=”img/portfolio/property1.jpg” class=”img-responsive” alt=”Project Title”> </a> </div>
     </div>
    </div>
    <div class=”col-sm-6 col-md-4 col-lg-4”>
     <div class=”portfolio-item”>
      <div class=”hover-bg”> <a href=”img/portfolio/02-large.jpg” title=”Project Title” data-lightbox-gallery=”gallery1”>
       <div class=”hover-text”>
        <h4>Adipiscing Elit</h4>
       </div>
       <img src=”img/portfolio/property4.jpg” class=”img-responsive” alt=”Project Title”> </a> </div>
     </div>
    </div>
```

```
<div class="col-sm-6 col-md-4 col-lg-4">
  <div class="portfolio-item">
    <div class="hover-bg"> <a href="img/portfolio/03-large.jpg" title="Project Title"
data-lightbox-gallery="gallery1">
      <div class="hover-text">
        <h4>Lorem Ipsum</h4>
      </div>
      <img src="img/portfolio/property8.jpg" class="img-responsive" alt="Project Title">
</a> </div>
  </div>
</div>
<div class="col-sm-6 col-md-4 col-lg-4">
  <div class="portfolio-item">
    <div class="hover-bg"> <a href="img/portfolio/04-large.jpg" title="Project Title"
data-lightbox-gallery="gallery1">
      <div class="hover-text">
        <h4>Lorem Ipsum</h4>
      </div>
      <img src="img/portfolio/property4.jpg" class="img-responsive" alt="Project Title">
</a> </div>
  </div>
</div>
<div class="col-sm-6 col-md-4 col-lg-4">
  <div class="portfolio-item">
    <div class="hover-bg"> <a href="img/portfolio/05-large.jpg" title="Project Title"
data-lightbox-gallery="gallery1">
      <div class="hover-text">
        <h4>Adipiscing Elit</h4>
      </div>
      <img src="img/portfolio/property3.jpg" class="img-responsive" alt="Project Title">
</a> </div>
  </div>
</div>
<div class="col-sm-6 col-md-4 col-lg-4">
  <div class="portfolio-item">
    <div class="hover-bg"> <a href="img/portfolio/06-large.jpg" title="Project Title"
data-lightbox-gallery="gallery1">
      <div class="hover-text">
        <h4>Dolor Sit</h4>
      </div>
      <img src="img/portfolio/property6.jpg" class="img-responsive" alt="Project Title">
</a> </div>
  </div>
</div>
<div class="col-sm-6 col-md-4 col-lg-4">
  <div class="portfolio-item">
    <div class="hover-bg"> <a href="img/portfolio/07-large.jpg" title="Project Title"
data-lightbox-gallery="gallery1">
      <div class="hover-text">
        <h4>Dolor Sit</h4>
```

```
          </div>
          <img src=”img/portfolio/property7.jpg” class=”img-responsive” alt=”Project Title”>
</a> </div>
        </div>
      </div>
      <div class=”col-sm-6 col-md-4 col-lg-4”>
        <div class=”portfolio-item”>
          <div class=”hover-bg”> <a href=”img/portfolio/08-large.jpg” title=”Project Title”
data-lightbox-gallery=”gallery1”>
          <div class=”hover-text”>
            <h4>Lorem Ipsum</h4>
          </div>
          <img src=”img/portfolio/property8.jpg” class=”img-responsive” alt=”Project Title”>
</a> </div>
        </div>
      </div>
      <div class=”col-sm-6 col-md-4 col-lg-4”>
        <div class=”portfolio-item”>
          <div class=”hover-bg”> <a href=”#” title=”Project Title” data-lightbox-
gallery=”gallery1”>
          <div class=”hover-text”>
            <h4>Adipiscing Elit</h4>
          </div>
          <img src=”img/portfolio/property9.jpg” class=”img-responsive” alt=”Project Title”>
</a> </div>
        </div>
      </div>
    </div>
  </div>
</div>
<!—Testimonials Section →
<div id=”testimonials”>
  <div class=”container”>
    <div class=”section-title”>
      <h2>Testimonials</h2>
    </div>
    <div class=”row”>
      <div class=”col-md-4”>
        <div class=”testimonial”>
          <div class=”testimonial-image”> <img src=”img/testimonials/01.jpg” alt=””> </div>
          <div class=”testimonial-content”>
            <p>”Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis sed dapibus leo nec
ornare diam sedasd commodo nibh ante facilisis 37ilename dolor feugiat at.”</p>
            <div class=”testimonial-meta”> - John Doe </div>
          </div>
        </div>
      </div>
      <div class=”col-md-4”>
        <div class=”testimonial”>
```

```
        <div class=”testimonial-image”> <img src=”img/testimonials/02.jpg” alt=””> </div>
        <div class=”testimonial-content”>
          <p>”Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis sed dapibus leo nec ornare diam sedasd commodo nibh ante facilisis.”</p>
          <div class=”testimonial-meta”> - Johnathan Doe </div>
        </div>
      </div>
    </div>
    <div class=”col-md-4”>
      <div class=”testimonial”>
        <div class=”testimonial-image”> <img src=”img/testimonials/03.jpg” alt=””> </div>
        <div class=”testimonial-content”>
          <p>”Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis sed dapibus leo nec ornare diam sedasd commodo nibh ante facilisis 38ilename dolor feugiat at.”</p>
          <div class=”testimonial-meta”> - John Doe </div>
        </div>
      </div>
    </div>
    <div class=”row”> </div>
    <div class=”col-md-4”>
      <div class=”testimonial”>
        <div class=”testimonial-image”> <img src=”img/testimonials/04.jpg” alt=””> </div>
        <div class=”testimonial-content”>
          <p>”Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis sed dapibus leo nec ornare diam sedasd commodo nibh ante facilisis 38ilename dolor feugiat at.”</p>
          <div class=”testimonial-meta”> - Johnathan Doe </div>
        </div>
      </div>
    </div>
    <div class=”col-md-4”>
      <div class=”testimonial”>
        <div class=”testimonial-image”> <img src=”img/testimonials/05.jpg” alt=””> </div>
        <div class=”testimonial-content”>
          <p>”Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis sed dapibus leo nec ornare diam sedasd commodo nibh ante facilisis 38ilename dolor feugiat at.”</p>
          <div class=”testimonial-meta”> - John Doe </div>
        </div>
      </div>
    </div>
    <div class=”col-md-4”>
      <div class=”testimonial”>
        <div class=”testimonial-image”> <img src=”img/testimonials/06.jpg” alt=””> </div>
        <div class=”testimonial-content”>
          <p>”Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis sed dapibus leo nec ornare diam sedasd commodo nibh ante facilisis.”</p>
          <div class=”testimonial-meta”> - Johnathan Doe </div>
        </div>
      </div>
    </div>
  </div>
```

```
  </div>
</div>
<!—Contact Section →
<div id=”contact”>
  <div class=”container”>
    <div class=”col-md-8”>
      <div class=”row”>
        <div class=”section-title”>
          <h2>Get In Touch</h2>
          <p>Please fill out the form below to send us an email and we will get back to you as soon as possible.</p>
        </div>
        <form name=”sentMessage” id=”contactForm” novalidate>
          <div class=”row”>
            <div class=”col-md-6”>
              <div class=”form-group”>
                <input type=”text” id=”name” class=”form-control” placeholder=”Name” required=”required”>
                <p class=”help-block text-danger”></p>
              </div>
            </div>
            <div class=”col-md-6”>
              <div class=”form-group”>
                <input type=”email” id=”email” class=”form-control” placeholder=”Email” required=”required”>
                <p class=”help-block text-danger”></p>
              </div>
            </div>
          </div>
          <div class=”form-group”>
            <textarea name=”message” id=”message” class=”form-control” rows=”4” placeholder=”Message” required></textarea>
            <p class=”help-block text-danger”></p>
          </div>
          <div id=”success”></div>
          <button type=”submit” class=”btn btn-custom btn-lg”>Send Message</button>
        </form>
      </div>
    </div>
    <div class=”col-md-3 col-md-offset-1 contact-info”>
      <div class=”contact-item”>
        <h4>Contact Info</h4>
        <p><span>Address</span>4321 California St,<br>
          San Francisco, CA 12345</p>
      </div>
      <div class=”contact-item”>
        <p><span>Phone</span> +1 123 456 1234</p>
      </div>
      <div class=”contact-item”>
        <p><span>Email</span> info@company.com</p>
```

```
    </div>
   </div>
   <div class="col-md-12">
    <div class="row">
     <div class="social">
      <ul>
       <li><a href="#"><I class="fa fa-facebook"></i></a></li>
       <li><a href="#"><I class="fa fa-twitter"></i></a></li>
       <li><a href="#"><I class="fa fa-google-plus"></i></a></li>
       <li><a href="#"><I class="fa fa-youtube"></i></a></li>
      </ul>
     </div>
    </div>
   </div>
  </div>
</div>
<script type="text/javascript" src="js/jquery.1.11.1.js"></script>
<script type="text/javascript" src="js/bootstrap.js"></script>
<script type="text/javascript" src="js/SmoothScroll.js"></script>
<script type="text/javascript" src="js/nivo-lightbox.js"></script>
<script type="text/javascript" src="js/jqBootstrapValidation.js"></script>
<script type="text/javascript" src="js/contact_me.js"></script>
<script type="text/javascript" src="js/main.js"></script>
</body>
</html>
```

Land Upload.java

```
import java.io.File;
import java.io.FileInputStream;
import java.io.FileOutputStream;
import java.io.IOException;
import java.io.InputStream;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;
import java.util.Iterator;
import java.util.List;
import java.util.logging.Level;
import java.util.logging.Logger;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
```

```
import javax.servlet.http.HttpSession;
import org.apache.tomcat.util.http.fileupload.FileItem;
import org.apache.tomcat.util.http.fileupload.FileUploadException;
import org.apache.tomcat.util.http.fileupload.disk.DiskFileItemFactory;
import org.apache.tomcat.util.http.fileupload.servlet.ServletFileUpload;

/**
 *
 * @author Admin1
 */
@WebServlet(urlPatterns = {"/LAND_UPLOAD"})
public class LAND_UPLOAD extends HttpServlet {

    /**
    * Processes requests for both HTTP
    * <code>GET</code> and
    * <code>POST</code> methods.
    *
    * @param request servlet request
    * @param response servlet response
    * @throws ServletException if a servlet-specific error occurs
    * @throws IOException if an I/O error occurs
    */
   protected void processRequest(HttpServletRequest request, HttpServletResponse response)
           throws ServletException, IOException, FileUploadException,
ClassNotFoundException, SQLException {
       response.setContentType("text/html;charset=UTF-8");
       PrintWriter out = response.getWriter();
       HttpSession session=request.getSession(true);

       try {
           String
O_Name="",O_Number="",O_Addr="",O_MAIL="",AREA="",CITY="",AMOUNT="",S
QFT = "",D_NO="",SUNO="",B_AMOUNT="";
           String saveFile="";
           int fileidnum=0,downloadcount=0,vc=0;
          // String contentType = request.getContentType();
           // Create a factory for disk-based file items
           DiskFileItemFactory factory = new DiskFileItemFactory();

// Set factory constraints
           factory.setSizeThreshold(4012);
//factory.setRepository("c:");

// Create a new file upload handler
           ServletFileUpload upload = new ServletFileUpload(factory);

// Set overall request size constraint
           //upload.setSizeMax(10024);
```

```
// Parse the request
        List items = null;
        try {
           items = upload.parseRequest(request);
        } catch (FileUploadException e) {
           e.printStackTrace();
        }
        byte[] data = null;
        String 42ilename = null;
// Process the uploaded items
        Iterator iter = items.iterator();
        while (iter.hasNext()) {
           FileItem item = (FileItem) iter.next();

           if (item.isFormField()) {
              //processFormField(item);

              String name = item.getFieldName();
              String value = item.getString();

              if (name.equalsIgnoreCase("O_Name")) {
                 O_Name = value;
                 System.out.println("O_Name" + O_Name);
              }

             else  if (name.equalsIgnoreCase("O_Number")) {
                 O_Number = value;
                 System.out.println("O_Number" + O_Number);
              }

             else  if (name.equalsIgnoreCase("O_Addr")) {
                 O_Addr = value;
                 System.out.println("O_Addr" + O_Addr);
              }
             else  if (name.equalsIgnoreCase("O_MAIL")) {
                 O_MAIL = value;
                 System.out.println("O_MAIL" + O_MAIL);
              }
             else  if (name.equalsIgnoreCase("AREA")) {
                 AREA = value;
                 System.out.println("AREA" + AREA);
              }
             else  if (name.equalsIgnoreCase("CITY")) {
                 CITY = value;
                 System.out.println("CITY" + CITY);
              }
              else  if (name.equalsIgnoreCase("AMOUNT")) {
                 AMOUNT = value;
                 System.out.println("AMOUNT" + AMOUNT);
              }
```

```
                else  if (name.equalsIgnoreCase("SQFT")) {
                    SQFT = value;
                    System.out.println("SQFT" + SQFT);
                 }
                else  if (name.equalsIgnoreCase("D_NO")) {
                    D_NO = value;
                    System.out.println("D_NO" + D_NO);
                 }
                  else  if (name.equalsIgnoreCase("SUNO")) {
                    SUNO = value;
                    System.out.println("SUNO" + SUNO);
                 }
                   else  if (name.equalsIgnoreCase("B_AMOUNT")) {
                    B_AMOUNT = value;
                    System.out.println("B_AMOUNT" + B_AMOUNT);
                 }
                 else {
                    System.out.println("ERROR");
                 }
               } else {
                  data = item.get();
                  43ilename = item.getName();
               }
            }
            saveFile = 43ilename;
                  String path1 = request.getSession().getServletContext().getRealPath("/");
              // String patt=path.replace("\\build", "");

               String strPath1 = path1+"\\"+saveFile;
            File ff1 = new File(strPath1);
            FileOutputStream fileOut1 = new FileOutputStream(ff1);
            fileOut1.write(data, 0, data.length);
            fileOut1.flush();
            fileOut1.close();
          out.println(saveFile);
FileInputStream fis11 = null;
File image1 = null;
//FileInputStream fis11 = null;
File image11 = null;
            Connection con7=null;
            PreparedStatement st7=null;
PreparedStatement st11=null;
image1 = new File(strPath1);
fis11 = new FileInputStream(image1);
    Class.forName("com.mysql.jdbc.Driver");
    con7 =
DriverManager.getConnection("jdbc:mysql://localhost:3306/property","root","admin");
String query="Select * from property_upload where D_NO='"+D_NO+"' or
SUNO='"+SUNO+"'";
```

```
System.out.println(query);
          Statement st=con7.createStatement();
ResultSet rs=st.executeQuery(query);

if(rs.next())
{

  session.setAttribute("msg","Already exist Please Check Values");
    response.sendRedirect("UPLOAD_PROPERTY.jsp");
}
else
{
      st7 =con7.prepareStatement("insert into property_upload values
(?,?,?,?,?,?,?,?,?,?,?,?,?,?)");

st7.setInt(1,0);
st7.setString(2,O_Name);
st7.setString(3,O_Number);
st7.setString(4,O_Addr);
st7.setString(5,O_MAIL);
st7.setString(6,AREA);
st7.setString(7,CITY);
st7.setString(8,AMOUNT);
st7.setString(9,SQFT);
st7.setString(10,D_NO);
st7.setString(11,SUNO);
st7.setString(12,B_AMOUNT);
if(44ilename != "")
      st7.setBinaryStream(13, (InputStream)fis11, (int)(image1.length()));
else
   st7.setBinaryStream(13, null);

    int I =st7.executeUpdate();
    session.setAttribute("msg","Successfully Upload");
    response.sendRedirect("UPLOAD_PROPERTY.jsp");
}

      }
      catch(Exception e)
      {
     out.println I;
      }
      finally {
         out.close();
      }
   }

   // <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on
the left to edit the code.">
   /**
```

```
     * Handles the HTTP
     * <code>GET</code> method.
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        try {
            processRequest(request, response);
        } catch (FileUploadException ex) {
            Logger.getLogger(LAND_UPLOAD.class.getName()).log(Level.SEVERE, null, ex);
        } catch (ClassNotFoundException ex) {
            Logger.getLogger(LAND_UPLOAD.class.getName()).log(Level.SEVERE, null, ex);
        } catch (SQLException ex) {
            Logger.getLogger(LAND_UPLOAD.class.getName()).log(Level.SEVERE, null, ex);
        }
    }
    /**
     * Handles the HTTP
     * <code>POST</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        try {
            processRequest(request, response);
        } catch (FileUploadException ex) {
            Logger.getLogger(LAND_UPLOAD.class.getName()).log(Level.SEVERE, null, ex);
        } catch (ClassNotFoundException ex) {
            Logger.getLogger(LAND_UPLOAD.class.getName()).log(Level.SEVERE, null, ex);
        } catch (SQLException ex) {
            Logger.getLogger(LAND_UPLOAD.class.getName()).log(Level.SEVERE, null, ex);
        }}
    /**
     * Returns a short description of the servlet.
     *
     * @return a String containing servlet description
     */
    @Override
    public String getServletInfo() {
        return "Short description";
    }// </editor-fold>}
```

PUBLICATIONS

ACCEPTANCE LETTER

ICBICC 2024-Full Paper Acceptance Notification-CC016 Inbox ×

ICBICC 2024 <info@icbicc.org> Fri, Mar 8, 11:13 AM

to me

Dear Shanmugapriya P, Namratha S R, Varunesh R V,

We are pleased to inform you that, after the double-blind peer review (please refer to the attached files), your manuscript identified above has been accepted for publication and oral presentation by 2024 2nd International Conference on Big Data, IoT, and Cloud Computing (ICBICC 2024) to be held in Xishuangbanna, on December 27-29, 2024.

Now, you may begin preparing for your presentation.

The preliminary program and location information will be posted on website: www.icbicc.org. Please compile and return the ICBICC Registration Form to info@icbicc.org. ICBICC 2024 conference committee will process confirmation when you are successfully registered and paid all the fees. You are suggested to finish the registration ***before March 22, 2024***.

All accepted and presented papers will be published in digital conference proceeding, which will send to be reviewed and indexed by major citation databases such as EI Compendex, Scopus, CPCI, Google Scholar etc.

Selected papers will be considered for submission in journal after significant extension. You will be given 15 minutes to present your paper during the conference and your presentation materials can be prepared in PDF or PPT format. The official language of ICBICC 2024 is English.

Thank you for your interest, we look forward to meeting you at ICBICC 2024!

RAJALAKSHMI ENGINEERING COLLEGE

DEPARTMENT OF CSBS

PROJECT WORK COURSE OUTCOME (COs):

CO1: On completion the students are capable of executing the proposed plan and become aware of and overcome the bottlenecks throughout every stage.

CO2: On completion of the project work students could be in a role to take on any difficult sensible issues and locate answers through formulating the right methodology.

CO3: Students will attain a hands-on level in changing a small novel idea / method right into an operating model / prototype related to multi-disciplinary abilities and / or understanding and operating in a team.

CO4: Students will be able to interpret the outcome of their project. Students will take on the challenges of teamwork, prepare a presentation in a professional manner and document all aspects of design work. '

CO5: Students will be able to publish or release the project to society.

RAJALAKSHMI ENGINEERING COLLEGE

DEPARTMENT OF CSBS

PROGRAM OUTCOMES (POs)

Engineering Graduates will be able to:

PO1 Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2 Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3 Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4 Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5 Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6 The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7 Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8 Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9 Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10 Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11 Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12 Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OBJECTIVES (PSOs)

A graduate of the Computer Science and Business Systems program will demonstrate:

PSO1:

The students graduating will have sound knowledge in Computer Science with equal appreciation of humanities, management sciences and human values.

PSO2:

The students will explore emerging topics such as Analytics, Machine Learning, Cloud Computing, and Internet of Things and apply the knowledge to design and develop solutions to societal needs.

PSO3:

The students will be industry ready with required business skills in service orientation.

CS19811 - PROJECT EVALUATION II

Project Title: PROPLED: Property Ledger

Batch Members: NAMRATHA S R (201401032)
VARUNESH R V (201401055)

Name of the Supervisor: Dr, P. Shanmugapriya, Professor,

Computer Science and Business Systems

CO-PO and PSO mapping

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO 9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO 1	2	2	3	2	2	3	2	2	1	2	2	2	2	3	2
CO 2	3	2	2	2	1	2	2	2	1	1	-	2	1	1	1
CO 3	3	1	2	3	1	2	2	2	1	1	-	2	2	1	2
CO 4	2	3	2	2	2	2	2	1	2	1	2	1	3	2	1
CO 5	2	2	2	1	2	1	2	1	-	2	-	2	2	1	2
Avg	2.4	2.0	2.2	2.0	1.6	2.0	2.0	1.6	1.0	1.4	0.8	1.8	2.0	1.6	1.6

Note: Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High), If there is no correlation, put -“

Signature of the Supervisor